

AWS Glue, Crawlers & Athena

Discover, Transform & Query Your Data

AWS Machine Learning Associate • Part 13

Presented by Prajan

What We'll Cover



The big picture



What is AWS Glue?



The Glue Data Catalog



Glue Crawlers & ETL jobs



Amazon Athena



When to use which

The Big Picture

1



Data in S3

CSV, JSON, Parquet files with no known structure

2



Glue Crawler

Scans the data and infers the schema

3



Data Catalog

Stores the table definitions

4



Amazon Athena

Query the files with plain SQL

Glue discovers & organizes. Athena queries.

What is AWS Glue?



Serverless data integration

- **Fully managed & serverless**
  No infrastructure to set up or maintain
- **Discover, prepare & move data**
  For analytics and machine learning
- **Pay for what you use**
  Charged for resources while jobs run

Data Catalog	Crawlers	ETL Jobs	Glue Studio

The four main pieces of Glue. We will go through each one.

The Glue Data Catalog



Databases

Organize your tables into logical groups



Tables

Record the schema (columns & types), the S3 location, and the file format



Metadata Only

The data itself stays in S3



Shared Across Services

Athena, EMR, Redshift Spectrum & Glue ETL all use the same definitions

One central place to describe all your data.

Glue Crawlers

1



Point at a Data Store

An S3 path, a JDBC database, or DynamoDB

2



Classify & Infer

Classifiers recognize the format and infer the schema

3



Update the Catalog

Creates or updates tables and detects partitions

4



Run & Repeat

On demand or on a schedule

New data and schema changes get picked up automatically.

Glue ETL Jobs



Serverless Spark Jobs

Write a PySpark or Scala script; Glue runs it on managed workers



Glue Studio

Build jobs visually by dragging and dropping, no code required



Triggers & Workflows

Run on a schedule or event, or chain several jobs together



Typical Tasks

Convert CSV to Parquet, clean data, join datasets, prepare ML features

ETL = Extract, Transform, Load.

Amazon Athena



SQL directly on S3

- **Interactive queries with standard SQL**
  Run them directly on data in S3
- **Serverless**
  Nothing to set up, just write a query
- **Pay per query**
  Charged by the amount of data scanned

Works with the Glue Data Catalog: tables come from the catalog, and query results are saved to an S3 location. Built on the open source Trino engine.

Making Athena Fast & Cheap



Use Columnar Formats

Parquet or ORC lets Athena read only the columns a query needs



Partition Your Data

For example by date, so queries skip folders they do not need



Compress Files

Fewer bytes to scan means lower cost



Avoid SELECT *

Name only the columns you want

You pay for data scanned, so scan less. A Glue job can convert CSV to partitioned Parquet.

When to Use Which

Discover the structure of data & add it to the catalog	**Glue Crawler**
Clean, convert or move data at scale, or on a schedule	**Glue ETL Job**
Quick, ad hoc SQL queries on data in S3	**Amazon Athena**
Table definitions shared across services	**Glue Data Catalog**

Recap

- Glue: a serverless service with a Data Catalog, crawlers & ETL jobs.
- Athena: SQL on data in S3, and you pay for what it scans.
- Together: from raw files to answers, with no servers to manage.

Thanks for watching!